

PC Assembly From Components

1. Install a CPU and a RAM stick onto a motherboard, then apply thermal paste using the pea-sized dot technique before mounting the CPU cooler.

Hint: Take a clear photo of each step as you complete it for your documentation.

->Here's a clear step-by-step procedure you can use for your practical documentation:

CPU, RAM & Cooler Installation

1. Prepare the motherboard

Place the motherboard on a clean, non-conductive surface and make sure the CPU socket and RAM slot are accessible.

2. Install the CPU

- Open the CPU socket retention mechanism.
- Align the small triangle/marker on the CPU with the matching mark on the socket.
- Gently place the CPU into the socket—**do not force it**.
- Secure the socket cover/retention arm.

3. Install the RAM stick

- Open the clips on the appropriate RAM slot.
- Align the notch on the RAM stick with the ridge in the slot.
- Press evenly on both ends until the clips lock into place.

4. Apply thermal paste

- If the cooler already has thermal paste pre-applied, don't add more.
- Otherwise, place **one small pea-sized dot** of thermal paste in the center of the CPU.
- Do not spread it manually unless the cooler manufacturer specifically recommends it.

5. Mount the CPU cooler

- Position the cooler directly over the CPU.
- Secure it using the manufacturer's mounting mechanism.
- Connect the cooler's fan cable to the motherboard's **CPU_FAN** header.

6. Take documentation photos

Take a clear photo after each major step:

-  CPU socket before installation
-  CPU correctly seated
-  RAM correctly installed
-  Pea-sized thermal paste on the CPU
-  CPU cooler mounted and fan connected

Safety: Work with the PC powered off and unplugged, discharge static electricity before handling components, and hold the CPU/RAM by their edges rather than touching the contacts.

2. Mount the motherboard inside a PC case and connect both the 24-pin ATX and 8-pin CPU power connectors from the power supply, making sure each connector clicks into place.

->Motherboard Mounting and Power Connection

1. Prepare the PC case

- Turn off and unplug the power supply.
- Remove the side panel.
- Check that the motherboard standoffs are positioned correctly.

2. Install the motherboard

- Carefully place the motherboard into the case.
- Align the rear I/O ports with the I/O opening.
- Align the mounting holes with the standoffs.
- Secure the motherboard with the appropriate screws. Do not overtighten.

3. Connect the 24-pin ATX power cable

- Locate the large **24-pin ATX connector** from the power supply.
- Align it with the motherboard's 24-pin socket.
- Press firmly until the locking clip **clicks** into place.

4. **Connect the 8-pin CPU power cable**

- Locate the **8-pin CPU/EPS power connector** from the power supply.
- Connect it to the CPU power socket, usually located near the top of the motherboard.
- Press until the locking clip clicks into place.

5. **Final check**

- Make sure both connectors are fully seated and their clips are locked.
- Ensure no cables are interfering with CPU fans or other components.

Documentation Photos

Take clear photos of:

- Motherboard mounted inside the case
- 24-pin ATX connector connected
- 8-pin CPU power connector connected
- Final cable connections

Important: Never force a connector. Make sure you are using the PSU's **CPU/EPS cable** for the CPU power socket, not a PCIe/GPU power cable.

3. Install a storage drive (SSD or HDD) and a GPU (graphics card) into the appropriate slots, then connect all necessary data and power cables to each device.

->**Storage Drive and GPU Installation**

1. **Install the SSD/HDD**

- Turn off and unplug the PC.
- For a **2.5-inch SATA SSD/HDD**, place the drive in the appropriate drive tray and secure it with screws.

- Connect a **SATA data cable** from the drive to a SATA port on the motherboard.
- Connect a **SATA power cable** from the power supply to the drive.
- For an **M.2 NVMe SSD**, insert it into the motherboard's M.2 slot at an angle, press it down, and secure it with the M.2 screw. No SATA data cable is required.

2. Install the GPU

- Locate the motherboard's main **PCIe x16 slot**.
- Remove the corresponding expansion-slot cover(s) from the rear of the case.
- Open the PCIe slot retention clip.
- Align the GPU with the slot and press it down evenly until it is fully seated and the clip locks.
- Secure the GPU to the case with the appropriate screw(s).

3. Connect GPU power

- If required, connect the appropriate **PCIe power cable(s)** from the power supply to the GPU.
- Make sure each connector is fully inserted and clicks into place.

4. Final check

- Verify that the storage drive's data and power connections are secure.
- Verify that the GPU is completely seated and securely mounted.
- Ensure cables are routed away from fans.

Documentation Photos

Take clear photos of:

- Storage drive mounted in the case
- SATA data and power connections **or** M.2 SSD installed
- GPU seated in the PCIe x16 slot
- GPU power cable connected
- Final completed installation

Safety: Never force a connector or component. Match each cable to the connector label and your motherboard/GPU manual.

4. Organize and secure all cables inside the case using zip ties or Velcro straps to ensure good airflow and a tidy appearance.

Hint: Take a photo showing your final cable management from both the front and back of the case.

->PC Cable Management

1. **Power off the PC** and unplug the power supply before organizing cables.
2. **Group the cables** by purpose:
 - 24-pin motherboard cable
 - 8-pin CPU power cable
 - GPU power cables
 - SATA power/data cables
 - Front-panel cables
 - Fan cables
3. **Route cables through the case cutouts** whenever possible. Keep them behind the motherboard tray to reduce clutter in the main chamber.
4. **Secure the cables** using zip ties or Velcro straps. Leave enough slack so connectors are not under tension.
5. **Keep cables away from fans** and avoid blocking the front-to-back airflow path.
6. **Check the final setup:**
 - Fans can spin freely.
 - No cables are touching hot components.
 - All important connectors remain firmly connected.
 - Side panels can close without excessive pressure.

Documentation Photos

Take two clear photos:

- **Front view:** Show the motherboard, GPU, storage, and neatly routed visible cables.
- **Back view:** Show the cables organized behind the motherboard tray using Velcro straps or zip ties.

Result: A well-organized cable layout improves airflow, makes troubleshooting easier, and gives the PC a clean appearance.

5. Perform a final verification: check for any loose screws, confirm all power/data connections are firmly attached, and document your completed build with a series of labeled photos showing each major component installed.

->Final PC Build Verification

1. Check for loose screws

- Inspect the motherboard, GPU, storage drive, and CPU cooler.
- Make sure all mounting screws are secure.
- Remove any loose screws left inside the case.

2. Check power connections

- Confirm the **24-pin ATX** motherboard connector is fully inserted.
- Confirm the **8-pin CPU/EPS** connector is secure.
- Check the **GPU power connector**, if required.
- Check SATA power connections for SATA drives.

3. Check data connections

- Confirm SATA data cables are firmly connected at both ends.
- If using an M.2 SSD, verify that it is properly seated and secured.

4. Check internal components

- CPU and cooler are securely installed.
- RAM sticks are fully seated with the slot clips locked.
- GPU is properly seated in the PCIe slot and secured to the case.
- All fans are connected to the appropriate fan headers.

5. Check cable management

- Ensure cables are secured with zip ties/Velcro.
- Make sure no cable is touching or blocking a fan.
- Confirm the side panels can be installed without forcing them.

6. Final visual inspection

- Look over the entire inside of the case once more.
- Confirm there are no tools, loose screws, packaging materials, or other objects left inside.

Labeled Documentation Photos

Take and label photos such as:

Photo	Label
 1	Motherboard Installed
 2	CPU & CPU Cooler Installed
 3	RAM Installed
 4	SSD/HDD Installed
 5	GPU Installed
 6	24-Pin ATX & 8-Pin CPU Power Connected
 7	Front Cable Management
 8	Back Cable Management
 9	Final Completed PC Build

Final result: The completed build should have all components securely mounted, all required power/data cables connected, no loose screws inside the case, and clean cable routing with unobstructed airflow.